NTNU

Corrective Source Term Approach

Hybrid analysis and modeling: learn what the solver's step is missing

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01

Motivation

Where physics-based models go wrong

A physics-based model (PBM) solves a governing equation we believe in — but the model we actually run is never exactly the system. Three things go wrong:

Unknown physics

Missing terms: an unmodeled source, sink, force or sub-grid effect.

Uncertain parameters

Coefficients (conductivity, damping, C_p) are wrong or drift.

Numerical error

A finite grid and time step approximate the operator.

The CoSTA idea in one line

Keep the solver; let a network supply a **corrective source term** inside each step, fitted so that the solver's step matches reference data.

Section 3 turns these three failure modes into four precise error cases. Benchmarks today: **heat** (PDE, full protocol) · **vessel** (ODE, transfer across maneuvers) · **wind turbine** (synthetic blade erosion).

Three ways to model, three ways to fail

	Known physics	Missing physics	Outside the training data
PBM	yes	no	bias persists; physics still applies
DDM	no	yes	the whole model is a guess
HAM / CoSTA	yes	as a correction	the correction is a guess

What the hybrid is for

The network only has to learn the *discrepancy*, usually a much smaller target than the whole dynamics.

What it does not buy automatically

Accuracy on new conditions, stability and conservation. Each must be checked: that is half of this lecture.

02

Formulation: the residual of the reference

Three objects

Reference u_{ref}

What we want to reproduce: a high-fidelity solver, an experiment, or a manufactured or synthetic "truth".

Runnable model (PBM)

$$\tilde{\mathscr{L}}\,\tilde{u} = \tilde{f}$$

the operator and source we have; in practice the solver's *discrete* step.

Corrected model

$$\tilde{\mathscr{L}}\,u_{\text{CoSTA}} = \tilde{f} + \sigma$$

the same operator with one extra source σ.

- Heat: $\tilde{\mathscr{L}} = \partial_t - D\,\partial_{xx}$ with Dirichlet data.
- Vessel ODE: $\tilde{\mathscr{L}} = \frac{d}{dt} - \boldsymbol{f}_{\text{PBM}}(\cdot)$ with an initial state.
- A "true" operator $\mathscr{L}$ may exist, but CoSTA never needs to evaluate it.

The exact corrective source

Substitute the reference into the corrected model and solve for σ:

Definition

$$\sigma^* = \tilde{\mathscr{L}}\, u_{\text{ref}} - \tilde{f} \qquad \text{(the residual of the reference in the runnable model)}$$

- With $\sigma = \sigma^*$ the reference solves the corrected problem. With matching initial/boundary data and a unique solution, $u_{\text{CoSTA}} = u_{\text{ref}}$.
- Built only from what we can evaluate: $\tilde{\mathscr{L}}, \tilde{f}$ and reference data. A *definition*, valid for any operator, linear or nonlinear.
- σ^* contains missing terms, wrong coefficients *and* the scheme's truncation error on this grid.

The catch

σ^* needs the reference, which we do not have when predicting. So we compute it on training data and learn $\sigma_{\boldsymbol{\theta}}$ from inputs available at prediction time.

Two cautions: a linear-only shortcut, and boundary data

"Operator applied to the error" is a shortcut

The definition $\sigma^* = \tilde{\mathscr{L}} u_{\text{ref}} - \tilde{f}$ always holds. Papers often rewrite it, using $\tilde{\mathscr{L}} \tilde{u} = \tilde{f}$, as

$$\sigma^* = \tilde{\mathscr{L}} u_{\text{ref}} - \tilde{\mathscr{L}} \tilde{u} \overset{\text{linear only}}{=} \tilde{\mathscr{L}} \left(u_{\text{ref}} - \tilde{u} \right).$$

The last step needs a *linear* $\tilde{\mathscr{L}}$. Logistic model: $\tilde{\mathscr{L}}(z_{\text{ref}} - \tilde{z}) = \sigma^* - 2\tilde{z}(z_{\text{ref}} - \tilde{z})$. Always compute the residual, never the shortcut.

Boundary data

σ acts inside the domain. If the model imposes a *wrong* Dirichlet value, the solver still imposes it whatever σ is: an interior source cannot repair boundary data.

03

The four error cases

Four ways the runnable model differs from the reference

If a true model $\mathscr{L}u = f$ exists, the exact label splits as

$$\sigma^* = \tilde{\mathscr{L}}_{\text{num}} u_{\text{ref}} - \tilde{f} = \underbrace{(\tilde{\mathscr{L}} - \mathscr{L})u_{\text{ref}}}_{\text{operator / parameter}} + \underbrace{(f - \tilde{f})}_{\text{source}} + \underbrace{(\tilde{\mathscr{L}}_{\text{num}} - \tilde{\mathscr{L}})u_{\text{ref}}}_{\text{truncation}}.$$

case	what is wrong	nonzero terms	demo (1D heat)
1	source or boundary data	source	forcing $0.8f$; wall at 1 instead of 1.2
2	parameter or operator term	operator	constant $\tilde{k}$ instead of $k(x)$
3	discretization only	truncation	exact physics, coarse grid
4	modelling *and* discretization	all	missing source on the coarse grid

Each demo: same trajectory split and protocol as the heat benchmark of Section 6, 3 seeds (`costa_error_cases.ipynb`).

Case 1: an inaccurate source

Derivation

$\tilde{\mathscr{L}} = \mathscr{L}$, same grid, $\tilde{f} \neq f$:

$$\sigma^* = \tilde{\mathscr{L}} u_{\text{ref}} - \tilde{f} = f - \tilde{f}.$$

Demo: $f = b\,4x(1-x)(1+0.5\sin 2\pi t)$, b varies by trajectory, $\tilde{f} = 0.8f$. The label equals $0.2f$ to 3×10^{-12}.

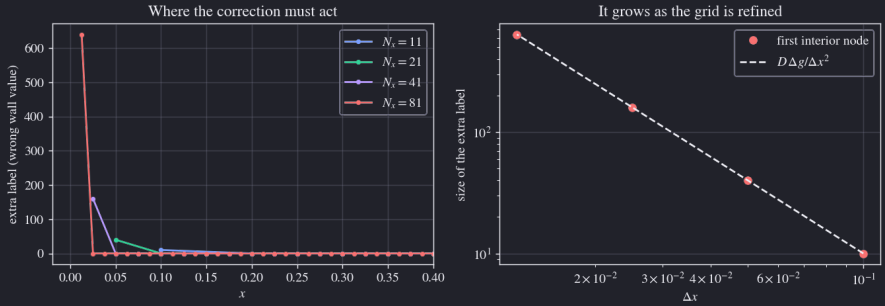
Test rollout RMSE (median of 3 seeds)

a (b)	PBM	stencil	stencil $+\tilde{f}$
1.25 (2.5)	6.4×10^{-2}	2.1×10^{-2}	6.8×10^{-5}
2.25 (1.0)	2.6×10^{-2}	2.1×10^{-2}	6.3×10^{-5}
3.75 (0.25)	6.4×10^{-3}	4.0×10^{-2}	8.5×10^{-4}

Lesson

Temperatures do not reveal b: the stencil-only network guesses an average source and can be worse than the PBM. The PBM's own forcing $\tilde{f}$ is known when predicting, so it is a legitimate input, and with it the error drops about $1000\times$.

Case 1, continued: inaccurate boundary data



Reference wall at $T = 1.2$, PBM wall at 1. The exact label exists, but it is a spike at the first interior node of size $D\Delta g/\Delta x^2$ ($10 \to 640$ as N_x goes $11 \to 81$), zero elsewhere, and the wall value itself stays wrong. Not a physical source: **fix boundary data at the boundary**. A wrong Neumann flux behaves like a source $\Delta q/\Delta x$ in the boundary cell, also grid-dependent.

Case 2: an inaccurate parameter

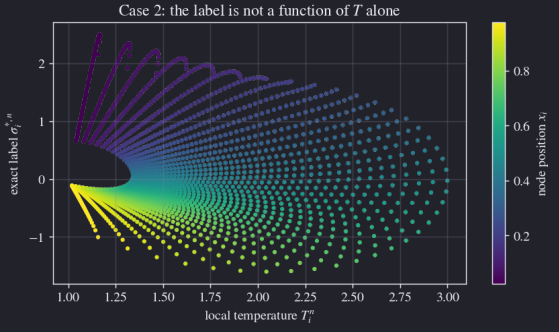
Derivation

$\tilde{\mathscr{L}} = \mathscr{L}(\tilde{k})$, same grid, $\tilde{f} = f$:

$$\sigma^* = (\tilde{\mathscr{L}} - \mathscr{L})u_{\text{ref}} = \partial_x\big((k - \tilde{k})\,\partial_x u_{\text{ref}}\big).$$

Linear $\tilde{\mathscr{L}}$ and $\tilde{\mathscr{L}}\tilde{u} = f$ give $\sigma^* = \tilde{\mathscr{L}}(u_{\text{ref}} - \tilde{u})$: the "operator applied to the error" form, valid here.

Demo: $k(x) = 0.1(1 + 2x)$, $\tilde{k} = 0.2$. Test RMSE ($a = 2.25$): PBM 8.7×10^{-2}, stencil 6.2×10^{-3}, stencil $+x$ 3.3×10^{-3}.



Case 3: discretization error only

Derivation

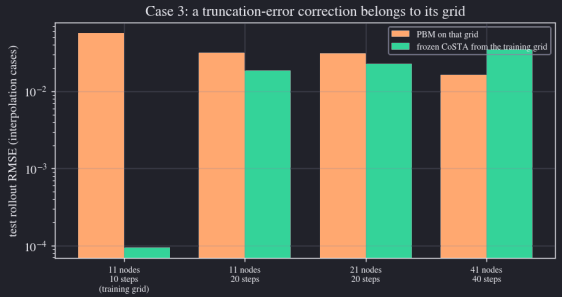
Exact physics and source, coarse scheme:

$$\sigma^* = (\tilde{\mathscr{L}}_{\text{num}} - \mathscr{L})u_{\text{ref}} = \text{truncation error.}$$

Demo: $T^* = 1 + a\sin(\pi x)e^{-D\pi^2 t}$ exactly, PBM on 11 nodes and 10 steps. Test RMSE $4 \times 10^{-2} \to 4 \times 10^{-5}$ on the training grid.

Lesson

The correction belongs to its grid. On a finer grid it is *worse* than the PBM.



Case 4: modelling and discretization together

Predictor–corrector (Blakseth et al.)

1. **Predictor**: $\tilde{\mathscr{L}}_{\text{num}}\left(\tilde{u}^{n+1}; u^n\right) = \tilde{f}$.
2. **Correction**: $\sigma_{\boldsymbol{\theta}}(\tilde{u}^{n+1}) \approx \sigma^{*,n}_{\text{num}}$, trained on labels built from the reference.
3. **Corrector**: $\tilde{\mathscr{L}}_{\text{num}}\left(u^{n+1}; u^n\right) = \tilde{f} + \sigma_{\boldsymbol{\theta}}$.

Demo: two-mode manufactured solution with a missing source, coarse grid. Label: missing source (mean $|s| = 5.5$) plus truncation (mean 1.1).

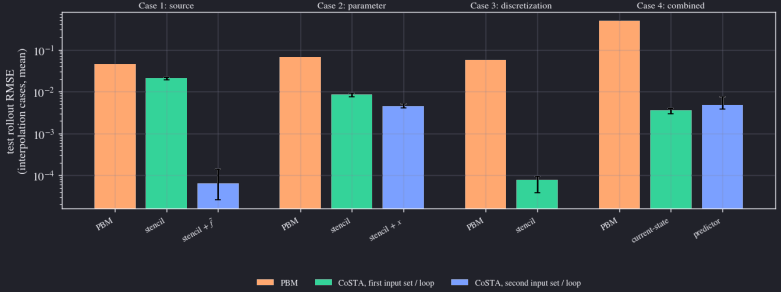
Test rollout RMSE (median of 3 seeds)

a	PBM	current-state	predictor
1.25	0.35	2.3×10^{-3}	2.3×10^{-3}
2.25	0.62	5.0×10^{-3}	7.3×10^{-3}
3.75	1.0	2.6×10^{-2}	4.4×10^{-2}

Finding

Both loops remove $> 99\,\%$ of the PBM error; the predictor costs two solves and is not better here.

The four cases side by side



case	label contains	inputs needed	valid on
1	missing forcing	stencil $+\tilde{f}$	any run whose forcing it sees
2	operator error	stencil $+x$	the training geometry
3	truncation	stencil	the training grid and step only
4	both	stencil	training grid, inside training states

04

Using the correction in a solver

The zero-correction property, stated precisely

What is true

With $\sigma_{\theta} \equiv 0$, one corrected step is exactly one PBM step **from the current state**. The PBM is always one switch away.

What does *not* follow

- "CoSTA is never worse than the PBM." A wrong nonzero correction can be worse, especially off the training data.
- "Switching it off undoes the error." It continues the PBM from the already drifted state.
- "The solver's stability and conservation carry over." They belong to the combined scheme, solver plus learned source, and must be checked.

Two prediction loops

Current-state (this course's notebooks): one solve per step



Predictor-based (Blakseth et al.): two solves per step



- Both keep the solver unchanged and feed the model's *own* state to the next step.
- Training input: the reference at level n (or the PBM step from it); label $\sigma^{*,n}_{\text{num}}$ from levels n and $n+1$.

05

Training, validation and testing

What a training pair may contain



- The **label** may use the future level $n+1$: it is built once, offline.
- The **input** may not: in a rollout the network sees the model's own state at level n.
- Training: minimize the MSE between $\sigma_{\boldsymbol{\theta}}(\boldsymbol{q}_i^n)$ and $\sigma_i^{*,n}$, with the solver held fixed — a plain regression.

Three roles for data

Training

Fits the weights *and* the input/label normalization.

Validation

Chooses the checkpoint, inputs, architecture, filter settings. Used repeatedly, so slightly optimistic.

Testing

Scores the frozen, chosen model *once*, on cases no decision used.

1. **Split whole trajectories first**, then extract rows. Neighbouring nodes and steps are near-copies; a random row split leaks.
2. **Normalize with training rows only.** A constant training channel needs a declared physical scale, not its (zero) standard deviation.
3. **Validate by rollout** and restore the best checkpoint.
4. **Test frozen and recursively**; touch the reference only to score.
5. **Report the spread** over declared seeds; record failed runs.

Three errors — report the right one

error	what is compared	answers
correction	σ_{θ} vs label, on reference states	did the regression fit?
one-step	one corrected step from a reference state vs next reference	is one step right?
rollout	recursive simulation on its own states vs reference	**does the simulation work?**

Why they differ

In a rollout the network is queried at slightly wrong states, where the training loss says nothing, and errors compound step by step.

Hence

Select checkpoints by *validation rollouts*; report *test rollouts*.

06

Heat benchmark: the complete protocol

Setup: what the PBM is missing

Reference (manufactured)

$$T^*(x,t;a) = 1 + a\sin(\pi x)\,e^{-t}$$

meets $T(0,t) = T(1,t) = 1$ for every a and solves $\partial_t T = D\,\partial_{xx} T + s$,

$$s = (D\pi^2 - 1)\,(T - 1).$$

PBM

$\partial_t T = D\,\partial_{xx} T$, backward Euler, 41 nodes, 50 steps to $t = 1$, $D = 0.5$. It omits s.

The missing physics is a reaction term

A heat source of strength $D\pi^2 - 1 \approx 3.93$ per unit excess temperature. The PBM's field decays to the boundary value far too fast. Keep this picture: it explains every result below.

Labels, and two checks before training

Exact discrete label (backward Euler)

$$\sigma_i^{*,n} = \frac{T_i^{*,n+1} - T_i^{*,n}}{\Delta t} - D\,\frac{T_{i+1}^{*,n+1} - 2T_i^{*,n+1} + T_{i-1}^{*,n+1}}{\Delta x^2}$$

Input: $\boldsymbol{q}_i^n = [T_{i-1}^n, T_i^n, T_{i+1}^n]$ only.

Check 1

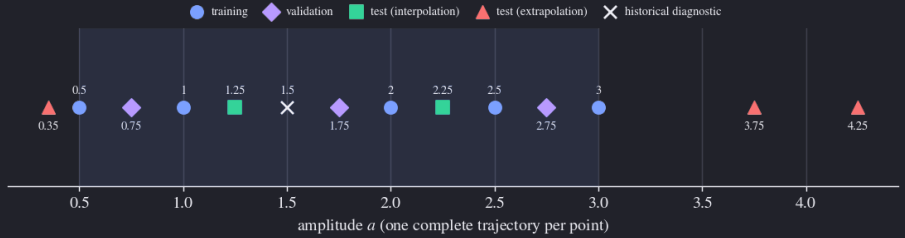
Exact label in one step $\Rightarrow$ next reference to 7×10^{-15}.

Check 2

Zero correction $\Rightarrow$ the PBM rollout, bit for bit.

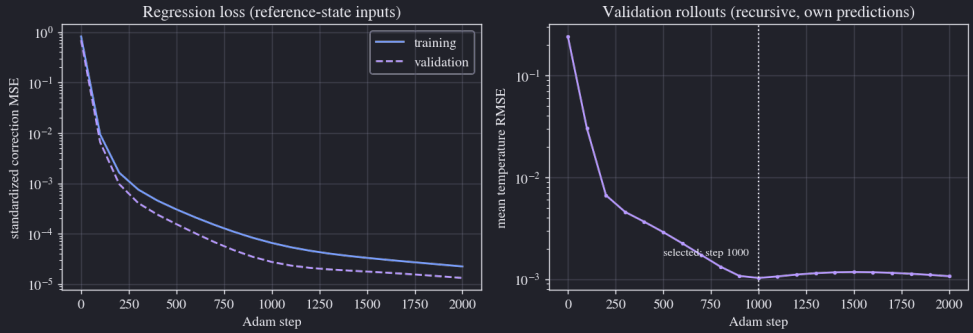
For this family the label is linear in the stencil, slope ≈ 3.85 vs. continuous 3.93: the gap is the scheme's truncation error (`costa_heat_pde.ipynb`).

The split: whole trajectories, fixed in advance



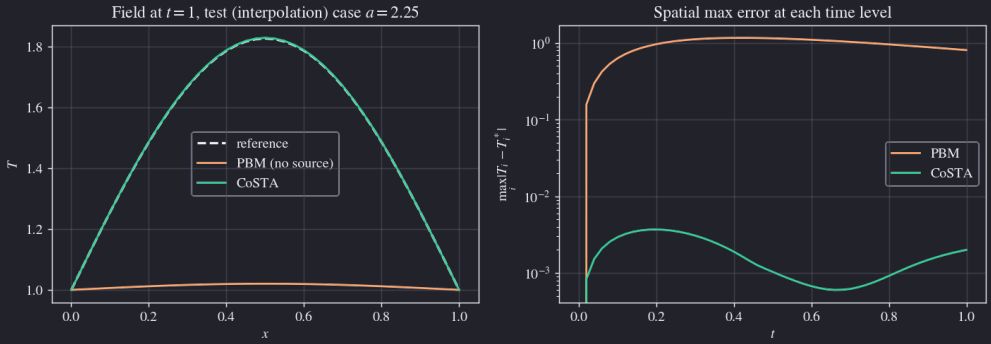
- Training $\{0.5, 1, 2, 2.5, 3\}$; validation $\{0.75, 1.75, 2.75\}$; test: interpolation $\{1.25, 2.25\}$, extrapolation $\{0.35, 3.75, 4.25\}$.
- $a = 1.5$ was inspected in earlier versions: a diagnostic only.
- MLP $3 \to 40 \to 40 \to 1$, Adam; validation rollouts every 100 steps; stop after 10 without a 1 % improvement.

Validation selects the checkpoint



Left: correction MSE on reference states keeps falling. *Right:* the validation rollout error levels off near 10^{-3} after $\sim$ 1000 steps; early stopping keeps step 1000. A better fit on reference states did not give a better simulation.

Test: a held-out amplitude, recursive rollout



Interpolation test case $a = 2.25$. *Left:* at $t = 1$ the PBM has decayed almost flat; CoSTA lies on the reference. *Right:* spatial max error at every time level. Rollout RMSE: PBM 0.68, CoSTA 1.3×10^{-3} (`costa_heat_pde.ipynb`).

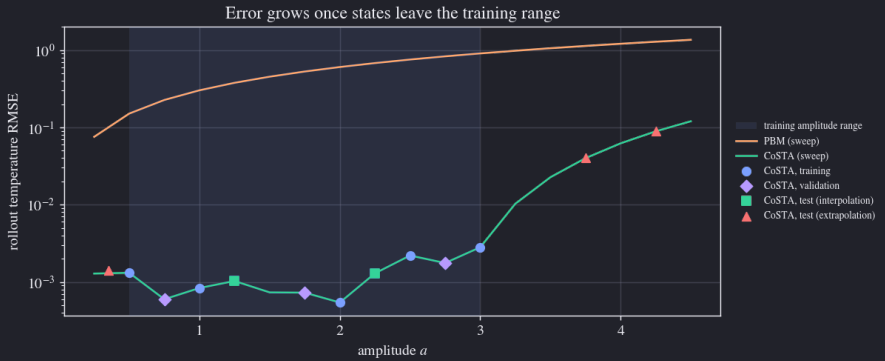
Test results, with their spread

role	a	PBM	CoSTA, seed 0	CoSTA, 5 seeds: median [range]
interpolation	1.25	0.38	1.0×10^{-3}	4.3×10^{-4} $[2.0 \times 10^{-4}, 1.0 \times 10^{-3}]$
interpolation	2.25	0.68	1.3×10^{-3}	9.4×10^{-4} $[2.3 \times 10^{-4}, 1.3 \times 10^{-3}]$
extrapolation	0.35	0.11	1.4×10^{-3}	1.1×10^{-3} $[4.0 \times 10^{-4}, 1.4 \times 10^{-3}]$
extrapolation	3.75	1.13	4.0×10^{-2}	3.2×10^{-2} $[1.8 \times 10^{-2}, 4.0 \times 10^{-2}]$
extrapolation	4.25	1.28	8.9×10^{-2}	7.5×10^{-2} $[5.0 \times 10^{-2}, 8.9 \times 10^{-2}]$

Rollout temperature RMSE over all nodes and time levels; all rollouts of all seeds finite.

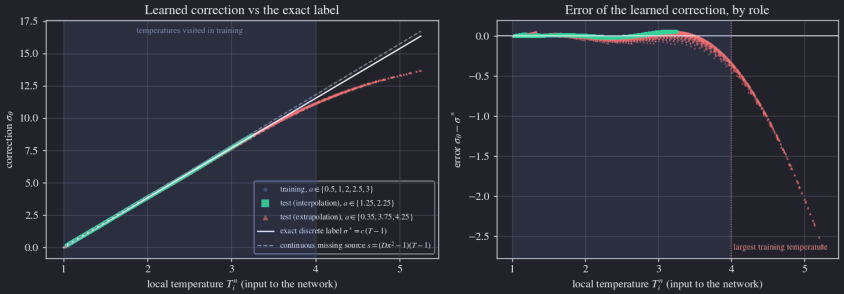
- In range: 350–500$\times$ below the PBM. Seed-to-seed: up to a factor of 5.
- $a = 0.35$ is “extrapolation” in a but behaves like interpolation. Why?

Across amplitudes



Lines: an exploratory sweep that mixes roles. Markers: the declared cases. CoSTA stays near 10^{-3} up to the top of the training range, then rises steeply; the PBM error grows with a because $s \propto a$.

New parameter $\neq$ new state

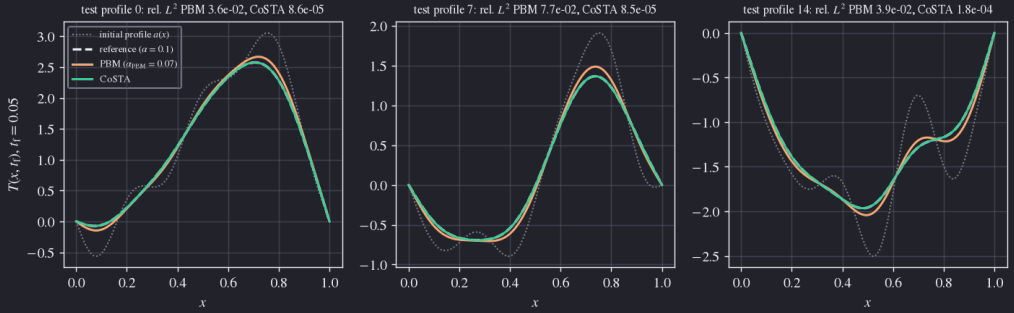


Left: the correction the network supplies during rollouts against the temperature it was given, by role; the exact label is a straight line. *Right:* its error against that label. The network sees **temperatures**, not a: it is exact wherever training temperatures exist and bends away above $T \approx 4$, reached only by $a = 3.75, 4.25$. The case $a = 0.35$ stays in $T \leq 1.35$, visited by every decaying training run, so it is *not* extrapolation.

Scope

One mode, one decay rate, one grid: transfer across a in this family only.

A second heat example: a wrong diffusivity (Case 2)



The course's canonical 1D rod (`heat1d`, used again in later lectures): $\alpha = 0.1$, eight-mode sine family, 128 points, FTCS to $t_f = 0.05$. PBM: the same march with $\alpha_{\mathrm{PBM}} = 0.07$ (Case 2). Split by profile: 60 training, 20 validation, 20 test. Relative L^2 error at t_f over the test profiles, median [max]: PBM 7.4×10^{-2} [0.14], CoSTA 9.8×10^{-5} [4.9×10^{-4}]; correction switched off = PBM bit for bit; Gaussian bump (out of family): PBM 0.13, CoSTA 4.6×10^{-3} (`costa_heat_rod.ipynb`).

07

Vessel benchmark: transfer across maneuvers

Setup: a missing drag, a residual force

Reference (synthetic)

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau} + \boldsymbol{d}(\boldsymbol{\nu})$$

$\boldsymbol{d} = [-3u|u|, -8v|v|, -2r|r|]^\top$, hidden from the models.

CoSTA

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}\boldsymbol{\nu} + \mathbf{D}\boldsymbol{\nu} = \boldsymbol{\tau} + f_{\boldsymbol{\theta}}(\boldsymbol{\nu})$$

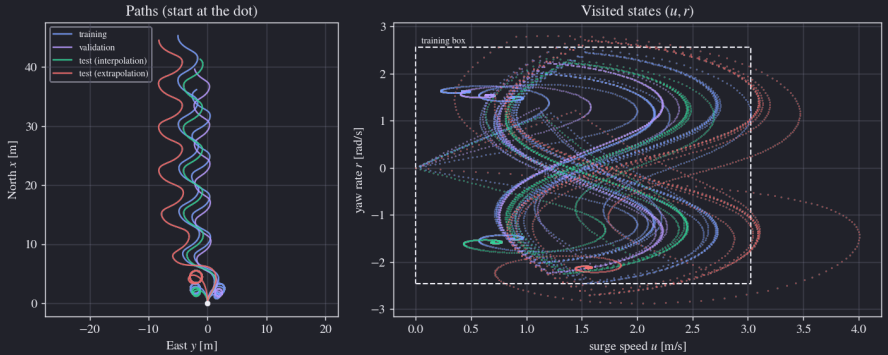
Input $\boldsymbol{\nu}$ suffices because $\boldsymbol{d}$ depends on $\boldsymbol{\nu}$ only.

Label: the residual force, one trajectory at a time

$$f_{\text{ref},k} = \mathbf{M}\dot{\boldsymbol{\nu}}_k + \mathbf{C}(\boldsymbol{\nu}_k)\boldsymbol{\nu}_k + \mathbf{D}(\boldsymbol{\nu}_k)\boldsymbol{\nu}_k - \boldsymbol{\tau}_k$$

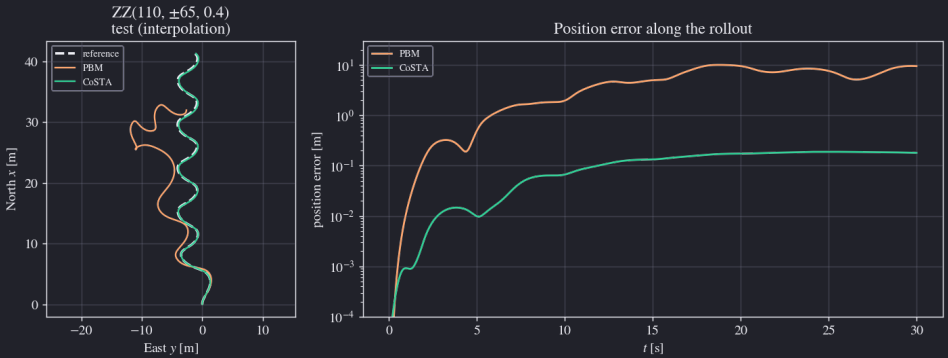
$\boldsymbol{\tau}_k$: the control *that* trajectory applied. $\dot{\boldsymbol{\nu}}_k$: central differences, never across a zig-zag switch (a difference across the jump is off by up to 77 N m).

Eleven maneuvers, split before anything is learned



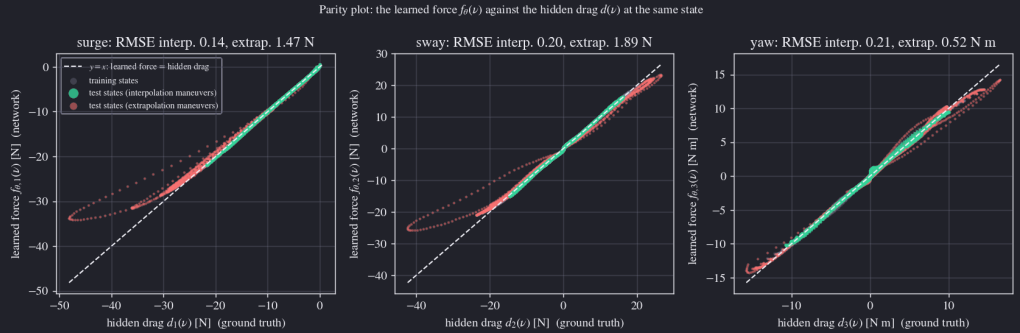
Training: 3 turning circles (both directions) and 2 zig-zags; validation: 2; test interpolation: TC$(110, -40)$, ZZ$(110, \pm 65)$, all states inside the training box; test extrapolation: ZZ$(160, \pm 80)$ (29 % of states outside) and TC$(300, -25)$ (94 % outside).

Held-out zig-zag (interpolation)



ZZ$(110, \pm 65, 0.4)$, frozen model, own predictions fed back. Position RMSE: PBM 5.8 m, CoSTA 0.13 m (5 seeds: 0.11–0.21 m). The RK4 solver does the transport; the network adds the force.

What was learned: the force on held-out states



Parity plot, one point per visited state. Horizontal axis: the hidden drag $\boldsymbol{d}(\boldsymbol{\nu})$ the reference applied at that state (ground truth, known only because the reference is synthetic). Vertical axis: the force the network produces at the *same* state. A point on the dashed $y = x$ line means the correction recovered the drag exactly. Training and interpolation-test states lie on the line (RMSE ≤ 0.21 N); extrapolation states bend away at the largest speeds.

A fair black box

DDM

$\dot{\boldsymbol{\nu}} = \boldsymbol{g}_{\boldsymbol{\phi}}(\boldsymbol{\nu}, \boldsymbol{\tau})$, MLP $6 \to 64 \to 64 \to 3$, no physics in the velocity dynamics; known kinematics.

CoSTA

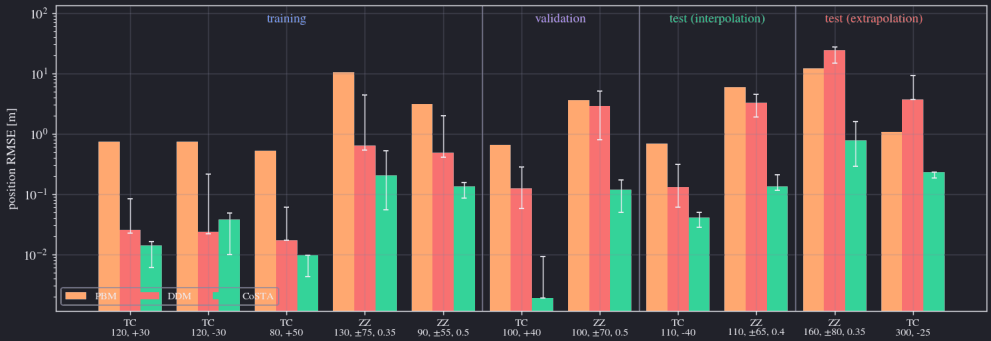
PBM $+ \mathbf{M}^{-1}\boldsymbol{f}_{\boldsymbol{\theta}}(\boldsymbol{\nu})$, MLP $3 \to 32 \to 32 \to 3$.

Identical for both:

- the same training samples and the same acceleration estimates;
- the same RK4 integrator as the reference, the same recorded controls;
- validation-selected checkpoints; inputs normalized with training statistics and declared physical floors (the old 10^{-9} denominator produced inputs of -6×10^{10}).

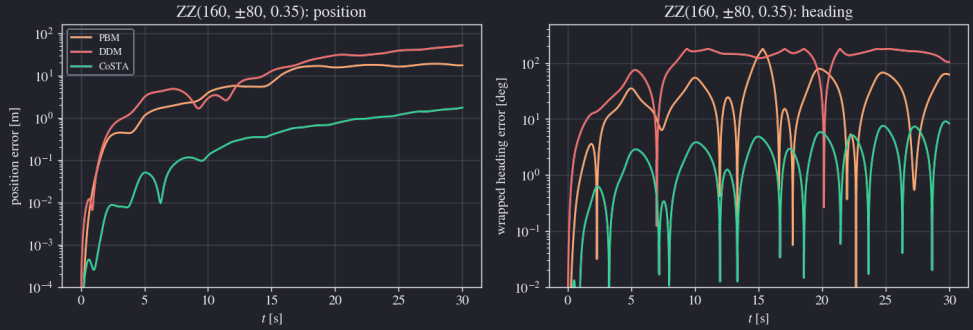
The only difference is the physics prior. The DDM was still improving when its 4000-step budget ran out.

PBM vs. DDM vs. CoSTA, by role



Position RMSE, seed 0, bars: range over 5 seeds. On training reconstructions DDM and CoSTA overlap. On every test maneuver CoSTA is lowest for every seed. The DDM beats the PBM in range but is *worse* than the PBM on both extrapolation maneuvers.

Outside the training range



ZZ(160, ±80, 0.35): position RMSE PBM 11.9 m, DDM 24.3 m, CoSTA 0.77 m (5 seeds: 0.29–1.6 m). Heading error is shown separately, wrapped, in degrees.

08

Wind turbine: synthetic blade erosion

Setup: a synthetic eroded turbine

Reference (simulated, not measured)

The course's NREL 5-MW rotor model, with a prescribed tip-erosion law:

$$T_{\text{aero}}^{\text{true}} = T_{\text{aero}}^{\text{nom}}(\omega, v, \beta)\,(1 - 0.015\,\lambda), \quad \lambda = \frac{\omega R}{v}.$$

$\approx 12\,\%$ torque loss at λ_{opt}.

CoSTA

$$J\dot{\omega} = T_{\text{aero}}^{\text{nom}} - T_{\text{gen}} + \sigma_{\boldsymbol{\theta}}(\omega, v, \beta)$$

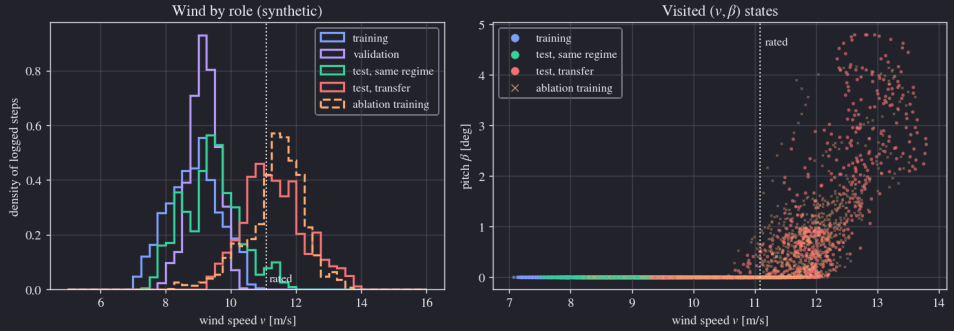
Pitch is an input: above rated the controller pitches the blades.

Labels from logged samples at the same step k

$$\sigma_k^* = J\,\frac{\omega_{k+1} - \omega_k}{\Delta t} + T_{\text{gen},k} - T_{\text{aero}}^{\text{nom}}(\omega_k, v_k, \beta_k)$$

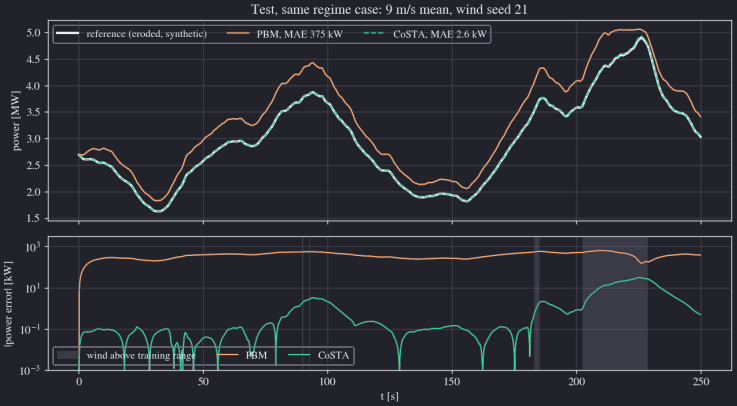
Equals the true deficit to 4×10^{-8} N m here (same explicit step as the reference); each rad/s of speed-log noise would cost ~ 546 kN m of label noise.

Wind realizations by role, and where the states go



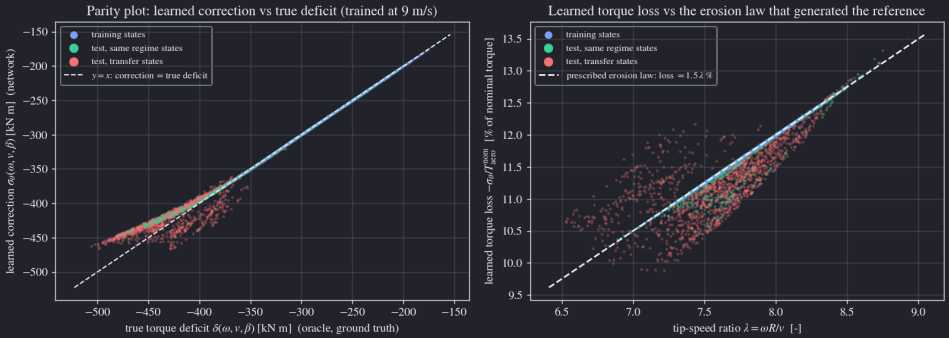
Training: 9 m/s mean, seeds 1–3; validation: seeds 11, 12; same-regime test: seeds 21, 22; transfer test: 11.5 m/s, seeds 31, 32 (above rated 11.1 m/s; pitch active in $\sim 40\,\%$ of steps, 83–99 % of states outside the training box). All training states have $\beta = 0$.

Same-regime test



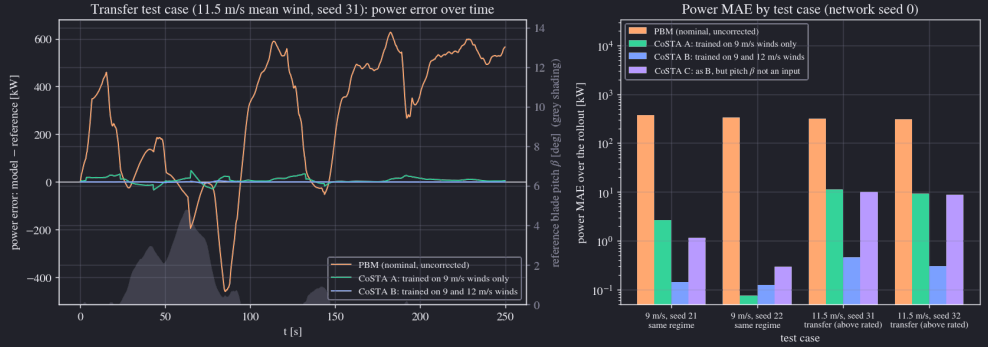
Power MAE, nominal → CoSTA: seed 22: 334 → 0.08 kW; seed 21 (shown): 375 → 2.6 kW, almost all of it during a gust above the training wind range (shaded).

What was learned: the torque correction against the true deficit



Left, parity plot: horizontal axis the *true* torque deficit at each state visited in a rollout (known because the reference is synthetic); vertical axis the network's correction at the same state. Training and same-regime states sit on $y = x$; transfer states miss by up to $\sim 50\,\text{kN m}$. *Right:* the same correction as a percentage of nominal torque against tip-speed ratio $\lambda = \omega R / v$; the dashed line is the erosion law $1.5\lambda\,\%$ that generated the reference, recovered where training data exist.

Transfer above rated



Three CoSTA variants, differing only in training data and inputs: **A** trained on the 9 m/s winds only; **B** trained on the 9 m/s winds *plus* two 12 m/s winds (seeds 41, 42); **C** as B but without the pitch angle β as a network input. *Left:* power error over time on transfer case 11.5 m/s, seed 31; grey shading is the reference blade pitch. *Right:* power MAE on every test case (network seed 0).

Transfer needs coverage *and* the right inputs

power MAE [kW], median of 5 seeds	11.5 m/s, seed 31	11.5 m/s, seed 32
nominal model (PBM)	317	307
CoSTA A: trained on 9 m/s winds only	12.5	9.7
CoSTA B: trained on 9 and 12 m/s winds	0.71	0.42
CoSTA C: as B, pitch β not an input	10.5	9.0

- Seed ranges of the three CoSTA variants do not overlap.
- The 9 m/s validation set would already have rejected variant C (0.23 vs. 0.08 kW for B).

Lesson

Transfer needed training data in the new regime *and* the input the missing torque depends on.

09

Properties, limits and outlook

Stability is a property of the corrected solver

Corrected backward-Euler heat step: $\mathbf{A}\boldsymbol{T}^{n+1} = \boldsymbol{T}^n + \Delta t\, \boldsymbol{\sigma}_{\boldsymbol{\theta}}(\boldsymbol{T}^n) + \boldsymbol{b}$, $\|\mathbf{A}^{-1}\|_2 = \frac{1}{1+\Delta t\, \lambda_{\min}}$.

Error recursion

$$\|\boldsymbol{e}^{n+1}\| \le \frac{1 + \Delta t\, L_{\boldsymbol{\theta}}}{1 + \Delta t\, \lambda_{\min}} \|\boldsymbol{e}^n\| + \frac{\Delta t\, \varepsilon}{1 + \Delta t\, \lambda_{\min}}$$

$L_{\boldsymbol{\theta}}$: network Lipschitz constant; ε: defect on reference states.

- Diffusion damps at rate $\lambda_{\min} \approx D\pi^2 \approx 4.9$; the network can amplify at rate $L_{\boldsymbol{\theta}}$. Bounded for all time if $L_{\boldsymbol{\theta}} < \lambda_{\min}$.
- Here the exact correction has slope 3.85: stable, with a margin a poorly regularized network could erase.
- Conservation: a free-form source adds net heat; a missing *flux* needs a flux-form correction.

What CoSTA gives, and its limits

Supported by the benchmarks

- Large rollout gains where test states resemble training states.
- A small learning target: only the discrepancy.
- An interpretable correction (a source, force or torque).
- The PBM one switch away (an identity, not a guarantee).

Limits

- Needs reference data; labels need derivatives or consecutive states.
- Inputs must expose what the correction depends on.
- States outside the training data: the correction is a guess.
- Rollouts can drift; stability must be checked.
- Same solver, grid and step in training and deployment.

The correction is data for discovery

The learned correction approximates the missing term on the visited states, denoised by the regression and evaluable on any grid of states we choose.

- A persistent, structured correction $\Rightarrow$ a genuine missing term.
- Its shape hints at the mechanism: a reaction term, a drag law, an erosion law.
- Next lecture (equation discovery): regress a sparse symbolic expression onto it.

Read with care

The label also contains truncation error, and different mechanisms can produce the same discrepancy on the visited states. A discovered term is a hypothesis to test on new data.

Checklist: what you must specify to evaluate CoSTA

Data and labels

- Which trajectories are training, validation, test?
- How are labels built (scheme inversion or derivatives, filter)?
- What may the input contain at prediction time?
- How is normalization fitted?

Evaluation

- What selects the checkpoint and the settings?
- Is the test a frozen, recursive rollout?
- Which metric, in which units?
- Are test states inside the training range?
- What is the spread over seeds, and did any run fail?

Check your understanding

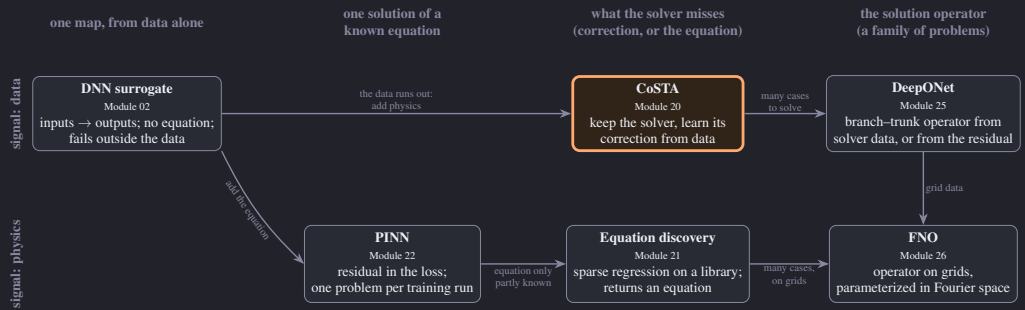
1. A paper reports a correction MSE of 10^{-6} on a random 20 % of rows. What kind of error is this, and what is wrong with the split?
2. You run a trained vessel correction with the reference velocity fed in at every step. One-step or rollout error?
3. Your test maneuver is new, but its (u, v, r) all lie inside the training range. Interpolation or extrapolation?
4. Is "with zero correction CoSTA is the PBM" a reason to expect CoSTA never to be worse than the PBM?

Takeaways

Takeaways

- Corrective source = residual of the reference in the runnable model.
- Labels may use the future; inputs may not. Select by validation rollouts, report frozen test rollouts.
- Gains are large where test states resemble training states; check stability and coverage, not just training loss.

Where this leaves us on the map



CoSTA keeps the solver and learns only the correction it needs, from data, and can switch the correction off. Its limitations: the correction is a black box and the solver stays in the loop. The arrows point to Module 21, which turns the correction into an equation, and to Modules 25 and 26, which learn the whole solution operator when many cases must be solved.

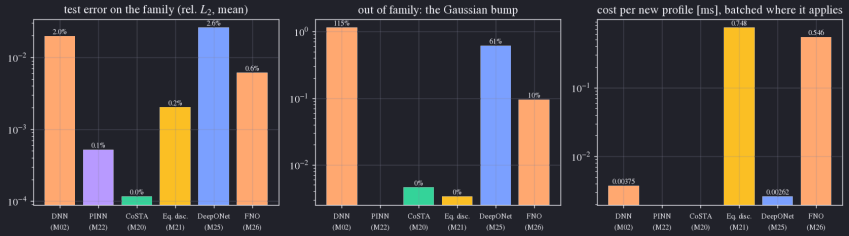


NTNU

Keep the physics — learn only what it misses, and test it honestly

Adil Rasheed · NTNU · adil.rasheed@ntnu.no

One rod, six methods: the numbers



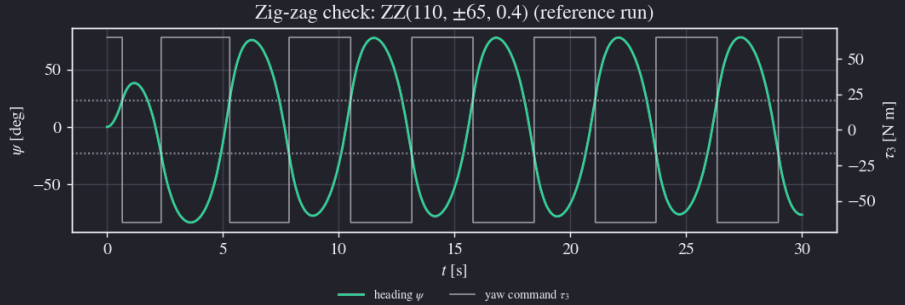
- Same rod ($\alpha = 0.1$, eight sine modes, FTCS truth), same test profiles, same metric: relative L_2 on $T(\cdot, t_\mathrm{f})$.
- **Left:** error on 50 unseen members of the family. **Middle:** one profile outside it (a Gaussian bump). **Right:** the cost of one new profile once the method exists (no PINN bar: a new profile is a new training run).
- A linear least-squares map is exact on this family, so the comparison is about the question each method answers, not about which is best.

One rod, six methods: what each one learned and what it cost

Method	What is learned	Physics needed	Solved cases	Test error	Bump	Cost / case	Training
DNN surrogate (Module 02)	40 sensor values $\to$ final profile	none	300	2.0%	115.0%	4 μs	9 s
PINN (Module 22)	one solution: $\sin \pi x$, $t \in [0, 1]$	the full equation	0	0.052%	–	retrain	per profile
CoSTA (Module 20)	the correction a wrong-α solver needs	an approximate solver ($\alpha = 0.07$)	60	0.012%	0.46%	solver + net per step	40 s
Equation discovery (Module 21)	α and the missing source $q(x)$	a library of candidate terms	20	0.2%	0.34%	0.75 ms	1 s
DeepONet (Module 25)	the operator $a \mapsto T(\cdot, t_f)$	none (data version)	300	2.6%	61.4%	3 μs	9 s
FNO (Module 26)	the operator $a \mapsto T(\cdot, t_f)$ on the grid	none (solver-made data)	300	0.61%	9.5%	0.55 ms	216 s

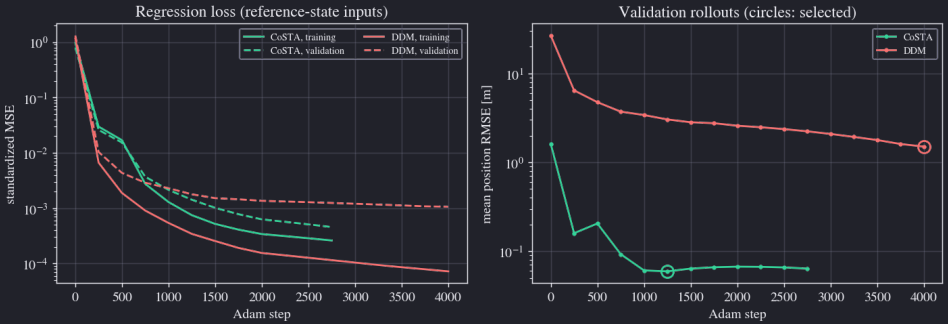
Solved cases: solver runs consumed for training. **Test error**: mean relative L_2 on 50 unseen profiles (PINN: its one profile, five seeds). **Bump**: the out-of-family profile. **Cost / case**: one new profile, batched where the method allows it; the FTCS solver itself takes about 0.6 ms per profile in a loop (Module 25 timing). Every number is recorded by the module's rod notebook.

Appendix: does the zig-zag zig-zag?



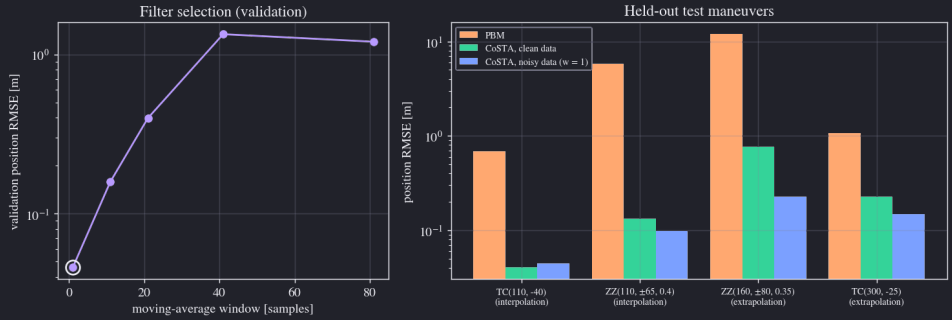
The command keeps its sign until the heading crosses the *opposite* switch line (hysteresis), and the yaw moment is strong enough to turn the hull back. The old law was memoryless and, at 30 N m, switched once and then turned continuously.

Appendix: vessel training curves



Validation selected CoSTA at step 1250 (early stop at 2750); the DDM at its last step, 4000.

Appendix: noisy labels, filter chosen on validation



Velocity noise std 0.004 m/s. Validation picks *no* smoothing ($w = 1$): the moving average biases the labels, the noise itself is zero-mean and the regression averages it. Test errors with noisy labels are within the seed spread of the clean-data model.

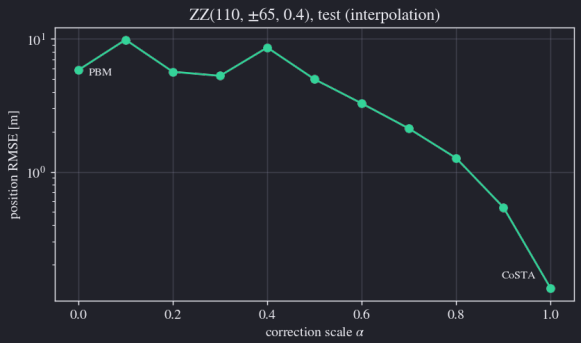
Appendix: should the heat network see x and t?

Earlier versions used $[T_{i-1}, T_i, T_{i+1}, x_i, t_n]$. Same protocol, five seeds each:

median rollout RMSE	**stencil only**	**stencil** $+x, t$
validation (decides)	7.3×10^{-4}	1.3×10^{-3}
test, interpolation $a = 2.25$	9.4×10^{-4}	4.5×10^{-4}
test, extrapolation $a = 4.25$	7.5×10^{-2}	9.1×10^{-2}
test, $a = 2.25$ to $t = 2$ (twice the horizon)	7.3×10^{-4}	3.8×10^{-2}

- Validation picks the stencil. Within the horizon the two are comparable.
- Past the training horizon the time input is new territory: $50\times$ worse. A state-only correction is also *autonomous*.

Appendix: scaling the correction is not a safety dial



Run ZZ$(110, \pm 65)$ with $\alpha f_{\boldsymbol{\theta}}$:

- $\alpha = 0$: the PBM, bit for bit.
- $\alpha = 0.1$ and 0.4: **worse than the PBM** (9.9, 8.6 m vs. 5.8 m).
- $\alpha = 1$: 0.13 m.

Lesson

A partial correction is a different model, not a safe compromise.